

PREDICTION OF SOIL MOISTURE USING MACHINE LEARNING MODEL

A Project Report

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CE 277: Remote Sensing in Ecohydrology

Submitted by

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1. Introduction

Soil moisture is a key variable in hydrology as it governs the partitioning of rainfall into infiltration, surface runoff, and evapotranspiration. It also plays a significant role in controlling land–atmosphere interactions, thereby influencing surface energy balance and local climatic conditions such as temperature and precipitation. Even small variations in soil moisture can lead to considerable changes in hydrological processes and ecosystem dynamics.

Global reanalysis datasets, such as ERA5, provide continuous and spatially extensive soil moisture information. However, these datasets often fail to accurately represent local-scale conditions due to coarse resolution and model assumptions. In contrast, in-situ observations, such as those from the AmeriFlux network, offer high accuracy but are limited in spatial coverage and may contain temporal gaps.

To address this limitation, this study explores the use of machine learning techniques to improve soil moisture estimation at the local scale. Specifically, a Random Forest regression model is developed to correct biases in ERA5 soil moisture using ground-based observations from the AmeriFlux site. The study aims to bridge the gap between large-scale reanalysis data and localized measurements, thereby enhancing the reliability of soil moisture predictions. Additionally, the model performance is evaluated using independent satellite observations from SMAP to assess its generalization capability.

2. Literature Review

2.1 Sungmin and Orth (2021)

This study developed a global-scale soil moisture dataset using a Long Short-Term Memory (LSTM) model trained on meteorological variables and in-situ observations. The model effectively captured temporal dynamics and provided soil moisture estimates across multiple depths. The study demonstrated that machine learning approaches can model complex relationships between climatic variables and soil moisture.

2.2 Wang et al. (2024)

Wang et al. presented a comprehensive review of deep learning techniques for soil moisture prediction. The study highlighted that soil moisture is highly nonlinear and influenced by multiple environmental factors. It emphasized that model performance depends strongly on input feature selection and temporal representation, with deep learning models performing particularly well for time-series data.

2.3 Tefera et al. (2026)

This study focused on integrating machine learning models with ground-based sensor data in agroecosystems. The results showed that combining in-situ measurements with meteorological variables significantly improves prediction accuracy. Models such as Random Forest, XGBoost, and LSTM achieved high performance ($R^2 > 0.9$). The study also identified temperature, soil heat flux, and evapotranspiration as key predictors.

2.4 Research Gap

Although previous studies demonstrate the effectiveness of machine learning in soil moisture prediction, most focus on large-scale modeling or specific datasets. There remains a gap in improving the accuracy of global reanalysis datasets, such as ERA5, at the local scale. Additionally, limited work has been done on validating such models using both ground-based observations and independent satellite datasets. This study aims to address these limitations.

3. Problem Statement

Accurate estimation of soil moisture is essential for understanding hydrological processes, agricultural planning, and climate interactions. Global reanalysis datasets such as ERA5 provide continuous and spatially extensive soil moisture information; however, they often exhibit significant bias and may not accurately represent local-scale conditions due to coarse spatial resolution and model assumptions.

On the other hand, ground-based observations, such as those from the AmeriFlux network, offer high accuracy but are limited in spatial coverage and may contain temporal gaps. This creates a gap between large-scale continuous datasets and accurate local observations, limiting the direct applicability of reanalysis data for site-specific studies.

Therefore, there is a need to develop a methodology that can reduce the bias in ERA5 soil moisture and improve its agreement with ground observations while maintaining continuous temporal coverage. This study addresses this challenge by applying a machine learning approach to enhance soil moisture prediction at the local scale.

4. Objectives

The objectives of this study are as follows:

- To develop a machine learning model for predicting soil moisture using ERA5 meteorological data.
- To reduce the bias present in ERA5 soil moisture estimates.
- To compare the model predictions with ground-based observations from the AmeriFlux site.
- To evaluate the performance of the model using independent satellite observations from SMAP.

5. Study Area

The study is conducted for the US-Var site (Vaira Ranch), which is part of the AmeriFlux network. The site is located in California, USA, at coordinates 38.4133° N latitude and 120.9508° W longitude, with an elevation of approximately 129 m above mean sea level.

The land cover of the region is predominantly grassland, making it suitable for studying surface soil moisture dynamics influenced by atmospheric conditions. The climate of the area is characterized by dry summers and wet winters, which leads to significant seasonal variation in soil moisture.

The site is particularly suitable for this study due to the availability of high-quality in-situ measurements and its sensitivity to seasonal climatic variations, making it ideal for evaluating soil moisture prediction models.

6. Data Description

This study utilizes data from three primary sources: ERA5 reanalysis data, AmeriFlux ground-based observations, and SMAP satellite data. All datasets correspond to the same study location (US-Var site) and are used for model development and validation.

6.1 ERA5 Reanalysis Data

ERA5 data is obtained from the European Centre for Medium-Range Weather Forecasts (ECMWF). It provides hourly meteorological variables with global coverage. The following variables are used in this study: air temperature (t2m), dew point temperature (d2m), precipitation (tp), surface solar radiation (ssrd), soil moisture (swvl1), and wind components (u10, v10). These variables serve as input features for the machine learning model. The variables are used in their respective standard units, such as temperature in Kelvin, precipitation in meters, solar radiation in J/m^2, and soil moisture in volumetric units (m^3/m^3).

6.2 AmeriFlux Data

AmeriFlux data is sourced from the AmeriFlux network for the US-Var flux tower site. It provides in-situ measurements of soil moisture with high accuracy. The soil moisture variable used in this study is SWC_PI_F_1_1_A, representing surface soil moisture. The original data is available at half-hourly resolution and is converted to hourly data to match the ERA5 dataset.

6.3 SMAP Satellite Data

SMAP (Soil Moisture Active Passive) data is obtained from NASA Earthdata. The Level 3 soil moisture product is used, which provides daily soil moisture estimates at a spatial resolution of 9 km. SMAP data is not used for model training but serves as an independent dataset for validating the performance and consistency of the machine learning model.

All datasets are pre-processed to ensure consistency in temporal resolution, units, and formatting. This step is essential for maintaining data integrity and ensuring accurate model training and evaluation.

7. Methodology

The methodology adopted in this study consists of data collection, preprocessing, alignment, model development, and evaluation. The overall workflow is designed to improve ERA5 soil moisture estimates using machine learning and validate the results with independent datasets.

7.1 Data Collection

Data is collected from three primary sources: ERA5 reanalysis data, AmeriFlux ground-based observations, and SMAP satellite data. ERA5 provides hourly meteorological variables, AmeriFlux provides in-situ soil moisture observations, and SMAP provides satellite-based soil moisture data for independent validation. All datasets correspond to the same study location (US-Var site).

7.2 Data Preprocessing

The collected datasets are pre-processed to ensure data quality and consistency. AmeriFlux data, originally available at half-hourly resolution, is converted to hourly data. Missing values and invalid data points are identified and removed. Relevant variables are selected from the

ERA5 dataset, and all datasets are standardized in terms of units and format. This step ensures that the data is suitable for machine learning model development.

7.3 Data Alignment

ERA5 and AmeriFlux datasets are aligned based on their timestamps to ensure temporal consistency. Only the time periods common to both datasets are retained. Any unmatched records are removed to avoid inconsistencies. This results in a clean dataset where input variables (ERA5) and target variable (AmeriFlux soil moisture) are properly paired for model training.

7.4 Model Development

A Random Forest regression model is used in this study due to its ability to capture nonlinear relationships and handle multiple input variables effectively. The input variables include air temperature (t2m), dew point temperature (d2m), precipitation (tp), solar radiation (ssrd), soil moisture (swvl1), and wind components (u10, v10). The target variable is the soil moisture observed at the AmeriFlux site.

The dataset is divided into training and testing sets using a 70:30 split. The model is trained on the training dataset to learn the relationship between ERA5 variables and observed soil moisture.

7.5 Model Evaluation and Validation

The performance of the model is evaluated using statistical metrics such as the coefficient of determination (R^2) and Root Mean Square Error (RMSE). The predicted soil moisture values are compared with AmeriFlux observations for both training and testing datasets.

To further assess model reliability, validation is performed using SMAP satellite data. The trends in soil moisture from the model predictions, ERA5 data, AmeriFlux observations, and SMAP data are compared to evaluate consistency and generalization capability.

The overall workflow involves preprocessing and aligning the datasets, followed by training the Random Forest model using ERA5 variables as inputs and AmeriFlux observations as targets. The trained model is then evaluated using statistical metrics and validated using independent SMAP data to assess its generalization capability.

8. Results and Discussion

The results obtained from the developed machine learning model are presented and analyzed in this section. The performance of the model is evaluated using both training and testing datasets, followed by comparison with ERA5 soil moisture and validation using SMAP satellite data.

8.1 Model Performance

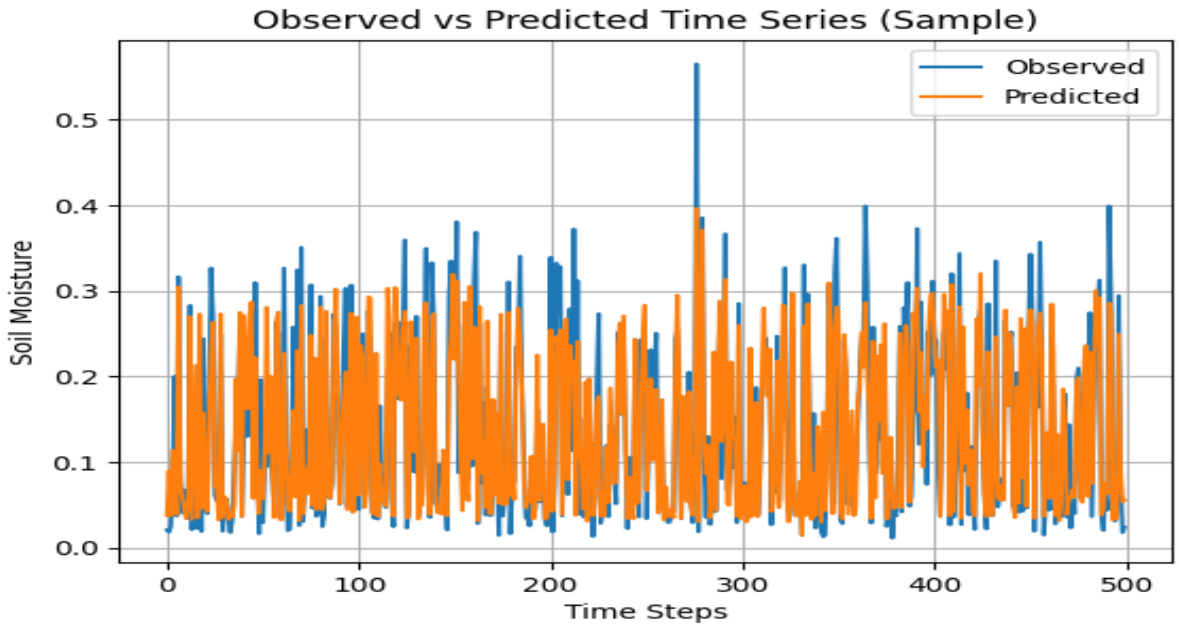


Figure 1: Time series comparison of observed and predicted soil moisture

Figure 1 shows the comparison between observed soil moisture (AmeriFlux) and the model predictions. It can be observed that the model is able to capture the general temporal variation in soil moisture reasonably well. The predicted values follow the overall pattern of the observed data, although some deviations are noticeable, particularly at higher soil moisture values. This indicates that while the model performs well under average conditions, it has limitations in accurately capturing extreme peaks.

Table 1: Performance of the machine learning model with ERA5 variables

Model	R^2	RMSE
Training	0.88	0.036
Testing	0.82	0.044

Table 1 shows that the model achieves strong performance, with a high coefficient of determination (R^2) and low RMSE values for both training and testing datasets. This indicates that the model is able to accurately capture the variability in soil moisture and generalize well to unseen data.

8.2 Comparison with ERA5 Soil Moisture

Table 2: Comparison of model performance with ERA5 soil moisture

Model	R^2	RMSE
Training	0.82	0.044
Testing	0.73	0.055

Table 2 demonstrates that the model performance decreases when only ERA5 soil moisture is used as input. This highlights the importance of incorporating additional meteorological variables to improve prediction accuracy. The full model significantly outperforms ERA5-only input, confirming the effectiveness of the machine learning approach in reducing bias.

ERA5 soil moisture (swvl1) shows a tendency to overestimate soil moisture values and exhibits limited variability over time. In contrast, the machine learning model reduces this bias significantly and produces predictions that are closer to AmeriFlux observations.

When the model is trained using only ERA5 soil moisture as input, the performance decreases ($R^2 = 0.73$, RMSE = 0.055), highlighting that additional meteorological variables play a crucial role in improving prediction accuracy.

8.3 Validation with SMAP Data

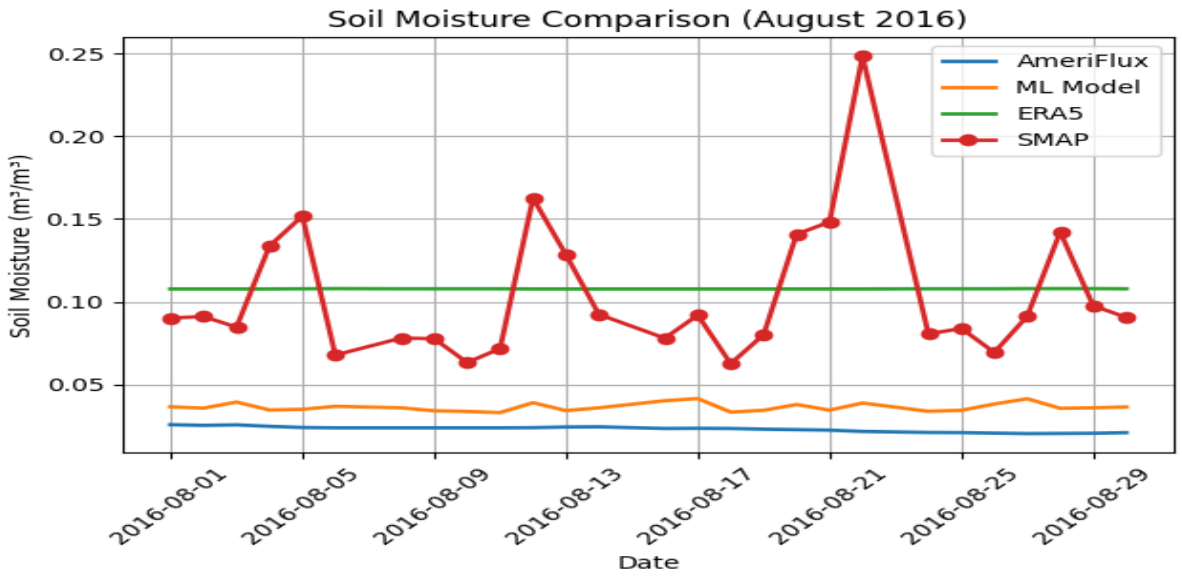


Figure 2: Comparison of soil moisture from AmeriFlux, ERA5, model predictions, and SMAP

Figure 2 shows the comparison between soil moisture obtained from AmeriFlux observations, ERA5 data, machine learning model predictions, and SMAP satellite data. It can be observed that ERA5 exhibits relatively constant values and tends to overestimate soil moisture. In contrast, SMAP shows higher variability due to its coarse spatial resolution. The machine learning model aligns more closely with AmeriFlux observations and captures the overall trend effectively. This indicates that the model improves soil moisture estimation while maintaining consistency with independent satellite observations.

8.4 Feature Importance Analysis

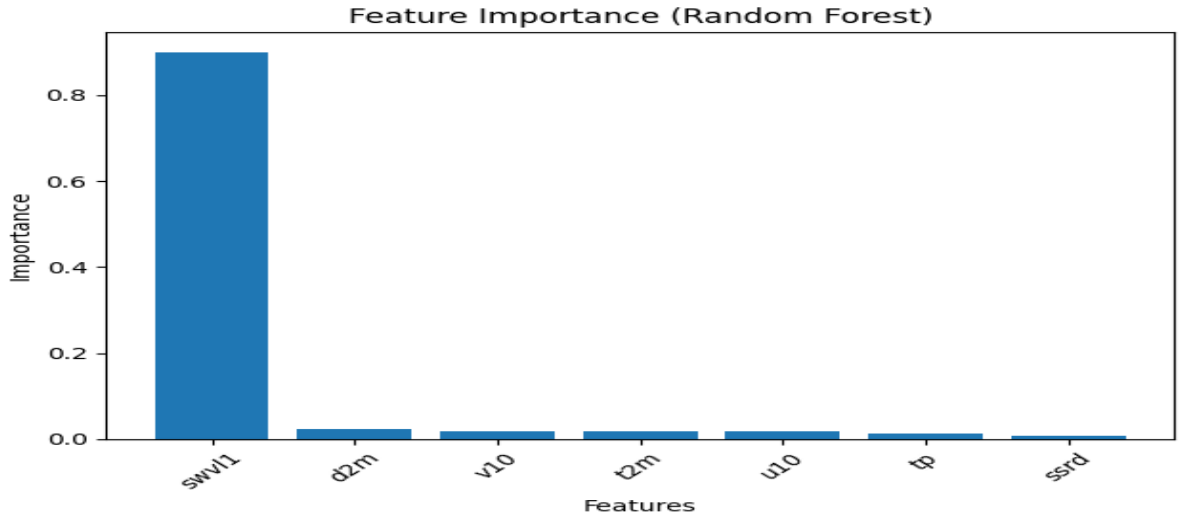


Figure 3: Feature importance of input variables in the Random Forest model

Figure 3 presents the relative importance of input variables in the Random Forest model. It is evident that ERA5 soil moisture (swvl1) is the dominant predictor, contributing the most to the model performance. This indicates that ERA5 provides a strong baseline estimate of soil moisture.

The remaining meteorological variables, including dew point temperature (d2m), air temperature (t2m), wind components (u10 and v10), precipitation (tp), and solar radiation (ssrd), have comparatively smaller contributions. However, these variables play a crucial role in improving the model predictions by capturing atmospheric influences and helping to correct the bias present in ERA5 soil moisture.

This result highlights that while the model heavily relies on ERA5 soil moisture, the inclusion of additional meteorological inputs enhances prediction accuracy and enables effective bias correction.

8.5 Overall Discussion

The results demonstrate that machine learning can effectively improve the accuracy of soil moisture estimates derived from reanalysis datasets. ERA5 soil moisture was found to exhibit systematic overestimation and limited variability, which restricts its direct applicability at the local scale.

The Random Forest model successfully reduces this bias by incorporating additional meteorological variables, leading to predictions that are more consistent with ground-based AmeriFlux observations. The time series analysis and validation with SMAP data indicate that the model captures the overall temporal trends of soil moisture reasonably well.

Feature importance analysis further reveals that while ERA5 soil moisture serves as a strong baseline predictor, other atmospheric variables contribute to improving model performance by capturing environmental influences. This highlights the importance of combining multiple data sources to enhance prediction accuracy.

However, the model shows some limitations in capturing extreme values, particularly at higher soil moisture levels. Despite this, the overall approach provides a reliable and practical method for improving local-scale soil moisture estimation.

9. Conclusion

This study demonstrates the effectiveness of using machine learning to improve soil moisture estimation at the local scale. The ERA5 dataset, although widely used, was found to overestimate soil moisture and exhibit limited variability, restricting its direct applicability for local-scale analysis.

The developed Random Forest model successfully reduces this bias and produces predictions that are closer to AmeriFlux observations. The inclusion of multiple meteorological variables significantly enhances prediction accuracy compared to using ERA5 soil moisture alone.

Validation with SMAP satellite data further confirms that the model is able to capture realistic soil moisture trends, indicating good generalization capability. Overall, the study highlights the potential of machine learning approaches in enhancing the reliability of soil moisture datasets, particularly in regions with limited ground observations.

10. Limitations

This study has certain limitations that should be acknowledged. The analysis is conducted for a single study site (US-Var), which limits the generalization of the model to other regions with different climatic and land cover conditions. Additionally, only surface soil moisture is considered, while deeper soil layers are not included in the analysis.

The study is based on a relatively short time period, which may not fully capture long-term variability and extreme events. Furthermore, the SMAP dataset used for validation has a coarse spatial resolution (9 km), which may not accurately represent local-scale soil moisture conditions.

Moreover, only one machine learning model (Random Forest) is implemented in this study, and the performance of other advanced models, such as deep learning approaches, is not explored.

11. Future Scope

The present study can be extended in several ways to further improve soil moisture prediction and its applicability. The model can be expanded to include deeper soil moisture layers, which would provide a more comprehensive understanding of subsurface hydrological processes.

Future work can involve applying the developed approach to multiple AmeriFlux sites with varying climatic and land cover conditions to improve model generalization. The incorporation of additional satellite datasets may further enhance validation and improve the robustness of the results.

Advanced machine learning and deep learning models, such as Long Short-Term Memory (LSTM) networks, can be explored to better capture temporal dependencies in soil moisture dynamics. Additionally, the development of spatially distributed models could enable regional-scale soil moisture estimation, thereby increasing the practical applicability of this approach in large-scale hydrological and agricultural applications.